1. Whole-system and individual-body analysis

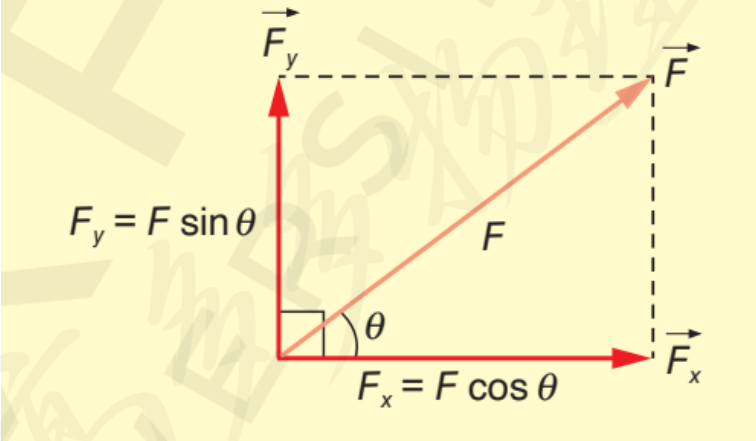
Objects with the same acceleration, for example those joined by an inextensible string, can first be treated as one system. Adding their force equations cancels internal tensions, allowing acceleration to be found from external forces and total mass. Then analyse an individual object to find tension or contact force.

2. Resolving a force into perpendicular components

A force can be replaced by perpendicular components with the same combined effect. If its angle is measured from the positive x-axis, the adjacent component uses cosine and the opposite component uses sine. Measuring from the vertical interchanges these roles. Assign signs according to the chosen axes and each component's direction.

$$F_x = F\cos\theta$$
$$F_y = F\sin\theta$$

F: force (N); θ: angle measured from the positive x-axis to F. Components are signed.



θ is measured from the horizontal component. The adjacent component is F cos θ and the opposite component is F sin θ.